

AgrIntelligence: An Explainable AI-Based Agricultural Decision Support System

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Abstract— Agriculture is influenced by several interconnected factors, including soil characteristics, crop conditions, weather variability, and fluctuations in agricultural market prices. Farmers often need to make decisions regarding crop productivity, market timing, and weather-related risks using fragmented information and conventional approaches. This paper presents AgriIntelligence, an explainable artificial intelligence based agricultural decision support system designed to integrate these factors into a unified, data driven platform. The proposed system collects crop and soil information, historical yield records, market price data, and real time weather information from application programming interfaces. After data preprocessing and feature engineering, machine learning and time series models are developed for crop yield prediction and market price forecasting. Multiple models are evaluated using appropriate prediction metrics and validation techniques to identify suitable models based on performance and interpretability. An explainable artificial intelligence layer is incorporated to provide insights into the factors influencing model predictions, thereby improving transparency and user trust. The system further integrates a real time data pipeline for continuous weather and market updates, enabling risk identification, alerts, and timely recommendations. A farmer centric dashboard presents predicted yield, market trends, weather conditions, agricultural risks, and recommended actions in an accessible form. Continuous monitoring, feedback collection, data quality assessment, and periodic model retraining enable the system to adapt to newly available information. The proposed framework aims to bridge the gap between predictive agricultural analytics and practical farm level decision making by providing transparent, real time, and actionable intelligence for improved agricultural planning and sustainable farming.

Keywords— Agricultural decision support system, artificial intelligence, machine learning, explainable artificial intelligence, crop yield prediction, market price prediction, real time weather analysis, precision agriculture.

I. INTRODUCTION

Agriculture plays a critical role in food security, economic development, and rural livelihoods. However, agricultural productivity is affected by several uncertain and interconnected factors, including soil characteristics, climatic conditions, crop selection, historical productivity, and market fluctuations. The increasing variability of weather conditions and uncertainty in agricultural markets make it difficult for farmers to make timely and informed decisions. Conventional decision-making practices frequently depend on historical experience and fragmented sources of information, which may not adequately capture the complex relationships among environmental, agronomic, and economic factors.

Recent developments in artificial intelligence, machine learning, deep learning, and data analytics have created opportunities for developing intelligent agricultural decision support systems. Machine learning models can analyze large volumes of agricultural data and identify relationships that may be difficult to determine using conventional approaches. Existing research has explored applications such as crop recommendation, soil classification, fertilizer recommendation, crop yield prediction, weather analysis, and agricultural price forecasting. For example, AI based decision support systems have combined soil, weather, crop health, and historical yield information to improve agricultural management decisions [4].

Crop yield prediction is particularly important because accurate yield estimates can support farm planning, resource allocation, and food supply management. Recent research has demonstrated the effectiveness of combining soil, weather, and remotely sensed information for yield prediction. AgriYieldAI, for example, integrates soil, time series weather, and satellite-derived information using a multi-branch deep learning architecture [5].

Another important challenge is agricultural market uncertainty. Crop prices can vary considerably because of seasonal patterns, weather conditions, production levels,

demand, and other market factors. Recent research has therefore investigated time series and hybrid machine learning approaches for crop price forecasting. Explainable approaches have also been introduced to identify the factors contributing to predicted price movements [2], [7].

Despite these developments, many agricultural AI systems focus on a specific prediction or recommendation task. Furthermore, highly accurate machine learning models may operate as black boxes, making it difficult for farmers and other stakeholders to understand why a particular prediction or recommendation was generated. Explainable artificial intelligence has consequently become an important component of trustworthy agricultural systems. Recent literature identifies transparency, interpretability, accountability, and stakeholder confidence as important considerations for AI adoption in agriculture [3].

To address these challenges, this paper proposes **AgriIntelligence**, an explainable AI based agricultural decision support system that combines agricultural, environmental, and economic information within a unified framework. The system integrates crop and soil data, historical yield information, market price records, and real time weather information. Machine learning and time series models are used to generate yield and market price predictions, while an explainable AI layer provides interpretable insights into model outputs.

The proposed system also incorporates a real time data pipeline that automatically retrieves updated weather and market information and processes the information for risk analysis. The resulting intelligence is presented through a farmer centric dashboard containing predicted yield, market trends, weather conditions, risk levels, and recommended actions.

The major objectives of the proposed system are:

1. To integrate heterogeneous agricultural, weather, and market data into a unified decision support framework.
2. To develop predictive models for crop yield and agricultural market prices.
3. To incorporate explainable AI for improving transparency and understanding of predictions.
4. To utilize real time weather and market information for dynamic agricultural risk analysis.
5. To provide actionable insights through a farmer centric dashboard.
6. To support continuous improvement through data quality monitoring, user feedback, and periodic model retraining.

II. RELATED WORK

Artificial intelligence has increasingly been applied to agricultural decision-making, with research covering crop recommendation, yield prediction, soil analysis, weather forecasting, fertilizer recommendation, and market price prediction.

Swati et al. proposed an agentic AI driven smart agriculture prediction system that integrates soil classification, soil parameter estimation, crop suggestion, fertilizer recommendation, and real time climatic information. Their system employed deep learning and ensemble learning techniques and incorporated explainability methods such as SHAP and LIME to improve transparency and user trust [1].

Patro et al. proposed EXACT-FARM, an explainable crop price forecasting framework based on a hybrid SARIMAX, XGBoost, and LSTM approach. The framework considers production trends, policy events, weather indices, and trade dynamics while using SHAP and counterfactual reasoning to explain price predictions [2].

About Samra and Ali reviewed the application of explainable artificial intelligence in smart agriculture and highlighted the importance of interpretability in agricultural AI systems. Their work discusses applications including crop classification, yield prediction, soil property assessment, irrigation management, and climate adaptation. The study identifies challenges such as heterogeneous data, scalability, uncertainty, computational complexity, and the need for trustworthy agricultural AI [3].

Suryawanshi et al. developed an AI based decision support system for precision agriculture by integrating agricultural information from multiple sources. The system considered soil moisture, weather conditions, crop health indices, and previous yield information and investigated models including Random Forest, Support Vector Machine, XGBoost, and LSTM. Their findings demonstrate the potential of AI based decision support for improving agricultural resource management [4].

Snehalatha and Patil introduced AgriYieldAI, which combines soil information, temporal weather data, and satellite imagery through YieldFusionNet. The system uses separate branches for different data sources and applies SHAP based feature importance to improve interpretability of yield predictions [5].

Trivedi et al. proposed a unified AI powered agricultural support system combining crop prediction, fertilizer recommendation, weather analysis, and farmer advisory services. The work emphasizes contextualized and real time recommendations across the farming lifecycle, particularly for small and marginal farmers [6].

Kumar et al. developed an explainable crop price prediction framework that integrates climatic and market dynamics. Their EHDL-SHAP framework combines recurrent neural networks for temporal climatic patterns with gradient boosted regression trees for non-temporal market information, while SHAP is used to identify the contribution of individual features [7].

Abishika et al. proposed an AI and machine learning based agricultural profitability and smart advisory system that combines soil image analysis with crop suitability, historical yield, market prices, and cultivation costs. The system focuses on generating economically relevant crop recommendations based on local conditions [8].

Marake et al. examined farmer centric crop recommendation and identified adaptability, explainability, context awareness, and usability as important requirements for agricultural AI systems. Their proposed framework emphasizes the need for transparent and context-sensitive crop recommendation systems [9].

III. Research Gap

The reviewed studies demonstrate significant progress in applying AI to individual agricultural problems. However, many existing systems focus primarily on a specific task such as crop recommendation, yield prediction, soil classification, or market price forecasting. Other systems integrate multiple agricultural functions but may not simultaneously combine **predictive analytics, real time data, market intelligence, explainability, and continuous model improvement** within a single farmer centric framework.

AgriIntelligence addresses this gap by integrating yield prediction, market price prediction, real time weather and market data, explainable AI, risk analysis, and farmer feedback within a unified decision support architecture.

IV. PROPOSED SYSTEM

A. AgriIntelligence is designed as an integrated agricultural intelligence platform that transforms raw agricultural data into predictions, explanations, risks, and actionable recommendations. The overall workflow consists of data collection, preprocessing, model development, evaluation, deployment, real time data processing, an AI insights layer, and continuous improvement.

- **System Architecture**

- The proposed architecture consists of eleven major stages.
- First, the system identifies the primary agricultural decision-making requirements, including crop yield estimation, weather risk monitoring, market price analysis, and simple explainable insights.

The major stakeholders considered are farmers, agricultural experts, and policymakers.

- The data collection layer gathers information from three primary sources: crop and soil datasets, market price datasets, and live application programming interfaces. The collected data is then processed and transformed into suitable features for machine learning models.
- The model building layer contains two primary prediction modules: a **Yield Prediction Model** and a **Market Price Prediction Model**. The yield model uses agricultural and environmental characteristics to estimate crop yield, while the market model uses historical price information and temporal features to estimate future prices.
- The prediction outputs are passed to an AI insights and decision layer, where the system combines model predictions with real time information and explainability mechanisms. Finally, the generated information is presented through a farmer dashboard.
- The architecture also includes continuous monitoring and feedback. Model performance, data quality, data drift, and farmer feedback are monitored so that models can be periodically retrained and improved.
- **[INSERT YOUR ARCHITECTURE FIGURE HERE]**
- **Fig. 1. Overall architecture of the proposed AgriIntelligence system.**

B. Data Collection

- The proposed system uses multiple sources of information to capture the agricultural conditions affecting farm-level decisions.
- **1) Crop and Soil Data**
- The crop and soil dataset contains information such as:
 - State and district
 - Crop
 - Crop area
 - Soil characteristics
 - Soil pH
 - Nitrogen
 - Phosphorus
 - Potassium
 - Historical yield
- Historical weather information, where available
- These variables provide information about the environmental and agronomic conditions associated with crop productivity.

- **2) Market Price Data**

- The market dataset contains agricultural commodity price information including:
 - Commodity
 - Market or mandi
 - Date
 - Minimum price
 - Maximum price
 - Modal price
 - Arrival or volume information
- Historical market data provides the basis for identifying temporal price patterns and generating future price predictions.

- **3) Real Time Data**

- Real time information is obtained using external application programming interfaces. Weather information is obtained through the Open-Meteo API, while agricultural market information can be obtained through government mandi or agricultural data sources such as data.gov.in and AGMARKNET.
- The use of real time data allows the system to supplement historical information with current environmental and market conditions.

C. Data Preprocessing

Raw agricultural datasets may contain missing values, duplicate observations, inconsistent formats, categorical variables, and outliers. Therefore, preprocessing is performed before model training.

The preprocessing pipeline includes:

1. Handling missing values.
2. Removing duplicate records.
3. Ensuring consistency in data formats and units.
4. Encoding categorical variables.
5. Scaling numerical features where required.
6. Detecting and handling outliers.
7. Performing feature engineering.

Feature engineering is particularly important for the market prediction model. Derived features may include lagged market prices, moving averages, price changes, seasonal information, month, year, weather aggregations, soil nutrient ratios, and district or state encodings.

These transformations enable the models to capture both temporal and contextual patterns in agricultural data.

D. Yield Prediction

The yield prediction component estimates expected crop productivity using agricultural, environmental, and historical information.

C. The input features may include:

- Location
- Crop
- Soil characteristics
- Weather conditions
- Historical yield
- Other engineered agricultural features

D. The target variable is the expected crop yield, represented in units such as kilograms per hectare or tonnes per hectare.

E. Candidate machine learning algorithms include:

- Random Forest
- XGBoost
- Gradient Boosting

F. Multiple models are trained and compared during evaluation. The model providing the most appropriate combination of predictive performance, robustness, and interpretability is selected for deployment.

G. This approach is consistent with previous research showing that ensemble learning methods can effectively model nonlinear relationships among agricultural variables [1], [4], [5].

H.

- **E. Market Price Prediction**

I. The market price prediction module forecasts future agricultural commodity prices based on historical market behavior and relevant contextual variables.

J. The input features include:

- Historical prices
- Lagged price values
- Moving averages
- Price changes
- Seasonal patterns
- Market information
- Weather related variables, where applicable

K. Candidate algorithms include:

- ARIMA
- SARIMA
- LSTM
- XGBoost
- LightGBM

L. Time series models are useful for capturing temporal dependencies, while tree based ensemble models can capture nonlinear relationships between market and contextual variables. Existing studies have demonstrated the value of combining temporal models with machine learning models for agricultural price forecasting [2], [7].

M. The target output is the future market price or a predicted price range.

F. EXPLAINABLE ARTIFICIAL INTELLIGENCE

One of the major components of AgriIntelligence is the explainability layer.

Conventional machine learning models may provide accurate predictions without explaining the factors responsible for those predictions. This can reduce user confidence, particularly in agricultural decision-making where farmers need to understand the reasoning behind recommendations.

Explainable artificial intelligence techniques can be used to identify the contribution of individual input features to a model prediction. For example, feature importance analysis can indicate whether soil properties, weather conditions, historical yield, or other variables had a significant influence on the predicted yield.

Similarly, for market price prediction, explainability can help identify the contribution of historical price patterns, seasonal variables, weather conditions, and other market factors.

This design follows the direction of recent agricultural research emphasizing SHAP, LIME, feature importance, and counterfactual reasoning for improving model transparency [1], [2], [3], [5], [7].

The explainability component therefore transforms a prediction from simply:

"Predicted yield: 4.8 tonnes per hectare"

into an interpretable output such as:

"The prediction is mainly influenced by historical yield, soil nutrient levels, and recent weather conditions."

This allows the farmer to understand the reasoning behind the generated insight.

G. REAL TIME DATA PIPELINE

AgriIntelligence incorporates a real time data pipeline to ensure that the decision support system does not rely exclusively on static historical datasets.

The pipeline operates on a daily basis and consists of two major streams.

1) 1) Weather Pipeline

The weather pipeline:

Open-Meteo API → Fetch current and forecast data → Store in database → Process data → Calculate risk → Generate alerts

The system can analyze parameters such as temperature, rainfall, humidity, and forecast conditions to identify potential agricultural risks.

2) 2) Market Pipeline

The market pipeline:

Market API → Fetch latest mandi prices → Store in database → Calculate trends and indicators → Detect sudden changes

The system can identify changes in market prices and generate market trend information for farmers.

The integration of real time information allows historical model predictions to be combined with current conditions, improving the practical usefulness of the decision support system.

V. IMPLEMENTATION

- A. *The AgriIntelligence system is implemented as a modular pipeline consisting of data ingestion, preprocessing, machine learning, explainability, backend services, database storage, and a farmer-facing dashboard.*
- B. **A. Data Processing Layer**
- C. *The data processing layer receives historical datasets and real time API information. Data cleaning, transformation, feature engineering, and validation are performed before the information is provided to the predictive models.*
- D. **B. Machine Learning Layer**
- E. *The machine learning layer contains separate models for yield prediction and market price prediction. Candidate models are trained using historical datasets and evaluated using appropriate validation strategies.*
- F. *For yield prediction, Random Forest, XGBoost, and Gradient Boosting can be compared.*
- G. *For market prediction, ARIMA, SARIMA, LSTM, XGBoost, and LightGBM can be evaluated depending on the characteristics of the dataset.*

C. Backend Layer

The backend provides an interface between the machine learning models, database, external APIs, and frontend dashboard.

The architecture specifies a backend based on frameworks such as **Flask** or **FastAPI**. The backend is responsible for:

Receiving user inputs.

Fetching real time data.

Calling trained prediction models.

Generating AI insights.

Providing prediction results to the frontend.

Managing database operations.

D. Database Layer

The database stores:

- Historical agricultural data.
- Market data.
- Weather information.
- Processed features.
- Prediction results.
- Model performance information.
- User feedback.

E. Dashboard Layer

The farmer dashboard presents the generated information in an accessible format.

The dashboard includes:

- Location
- Crop
- Current weather
- Rainfall
- Humidity
- Weather risk
- Current market price
- Market change
- Market trend

- Estimated yield
- Recommended actions

VI. RESULTS AND EVALUATION

Important: This section is the one place where I need your actual experimental results before we claim specific numbers. Your architecture tells us what should be evaluated, but it does not contain your actual model results. So for now, use this structure.

A. Yield Prediction Evaluation

The yield prediction models are evaluated using regression metrics including:

Root Mean Square Error

Mean Absolute Error

R squared score

Cross-validation performance

The models are compared to determine which approach provides the most reliable yield predictions.

TABLE I. YIELD PREDICTION MODEL PERFORMANCE

Model	RMSE	MAE	R ²
Random Forest	[VALUE]	[VALUE]	[VALUE]
XGBoost	[VALUE]	[VALUE]	[VALUE]
Gradient Boosting	[VALUE]	[VALUE]	[VALUE]

Based on the experimental results, **[BEST MODEL]** achieved the best overall performance with an RMSE of **[VALUE]**, MAE of **[VALUE]**, and R² of **[VALUE]**.

B. Market Price Prediction Evaluation

Market price prediction models are evaluated using:

RMSE

MAE

Mean Absolute Percentage Error

Time based validation

Backtesting

TABLE II. MARKET PRICE PREDICTION MODEL PERFORMANCE

Model	RMSE	MAE	MAPE
ARIMA	[VALUE]	[VALUE]	[VALUE]
SARIMA	[VALUE]	[VALUE]	[VALUE]
LSTM	[VALUE]	[VALUE]	[VALUE]
XGBoost	[VALUE]	[VALUE]	[VALUE]
LightGBM	[VALUE]	[VALUE]	[VALUE]

The model with the most suitable performance is selected for deployment based on predictive accuracy and interpretability.

C. Explainability Evaluation

The explainability component is evaluated based on whether the system can provide meaningful feature-level explanations for individual predictions.

For yield prediction, important features may include:

Soil nutrient levels

Soil pH

Historical yield

Weather variables

Crop and location

For market prediction, important features may include:

Previous prices

Moving averages

Seasonal factors

Market conditions
Weather related variables

The feature importance results can be visualized using SHAP based plots or other appropriate explainability techniques.

[INSERT SHAP/FEATURE IMPORTANCE FIGURE HERE]

Fig. 2. Feature importance for the yield prediction model.

D. Real Time System Evaluation

The real time pipeline is evaluated based on:

Data retrieval success.
Data freshness.
API response time.
Data processing time.
Prediction generation time.
Alert generation.
Dashboard update frequency.

The system is designed to automatically update weather and market information so that users can access current agricultural conditions.

VII. AI INSIGHTS AND FARMER DECISION SUPPORT

The primary purpose of AgriIntelligence is not only to produce predictions but also to convert predictions into understandable agricultural insights.

The AI insights layer combines model outputs with current weather and market information to produce actionable information.

For example, the system can present:

A. Predicted Yield

The system provides an estimated yield range based on crop, location, soil, weather, and historical information.

Example:

Estimated Yield: 4.5 to 5.1 tonnes per hectare

B. Market Trend

The system displays the current market price together with its recent change and predicted trend.

Example:

Today's Price: 2750

Change: Increasing

C. Weather Risk

Current and forecast weather data are processed to identify potential risks.

Example:

Weather Risk: Moderate

This allows farmers to consider weather conditions while making operational decisions.

D. Agricultural Recommendations

The system combines the predictions and risk analysis to provide simple recommendations.

For example:

Consider selling if the predicted market trend is favorable.
Monitor weather conditions before undertaking weather-sensitive activities.
Review expected yield before planning harvest and storage.
Consider the model's important contributing factors before acting on a prediction.

E. Explainable Recommendations

Instead of simply displaying a recommendation, the system provides an explanation for the generated result.

For example:

Recommendation: Monitor weather conditions during the upcoming week.

Reason: The recommendation is influenced by the predicted rainfall and associated weather risk.

This farmer-centric approach is consistent with recent research emphasizing contextualized, interpretable, and usable agricultural recommendations [6], [9]

VIII. CONCLUSION AND FUTURE WORK

A. A. Conclusion

This paper presented AgriIntelligence, an explainable AI based agricultural decision support system designed to integrate agricultural, environmental, and market information into a unified platform. The proposed framework combines crop and soil data, historical yield information, market price records, and real time weather information to support farm-level decision-making.

Machine learning and time series approaches are incorporated for crop yield prediction and market price forecasting, while an explainable AI layer provides insights into the factors influencing model predictions. The integration of real time weather and market data enables the system to move beyond static prediction and provide dynamic information regarding weather risks and market trends.

The proposed farmer-centric dashboard converts model outputs into accessible insights including estimated yield, market conditions, weather status, risk levels, and recommendations. In addition, continuous monitoring, user feedback, data quality assessment, and periodic model retraining provide a mechanism for improving the system as new data becomes available.

By combining predictive analytics, real time information, explainability, and continuous learning, AgriIntelligence aims to bridge the gap between advanced artificial intelligence techniques and practical agricultural decision-making. The framework can support farmers and other agricultural stakeholders in making more informed decisions while encouraging efficient resource utilization and sustainable farming practices.

B. B. Future Work

Future development of AgriIntelligence can focus on several areas.

First, the system can be extended with additional data sources such as satellite imagery, Internet of Things sensors, crop health indices, and soil images. These additional sources can improve the system's ability to capture field-level variations.

Second, advanced deep learning and hybrid models can be investigated to improve yield and market price prediction while maintaining interpretability.

Third, the system can be extended to provide multilingual and voice-based interaction to improve accessibility for farmers with limited technical literacy.

Fourth, uncertainty estimation can be incorporated so that the system communicates prediction confidence alongside its outputs.

Finally, continuous learning mechanisms can be improved by incorporating additional farmer feedback and newly observed agricultural outcomes, allowing the models to adapt to changing environmental and market conditions.

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